THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The prime factorisation of 140 is: 1

(a) $2 \times 5 \times 7$ **(b)** $2^3 \times 5 \times 7$ **(c)** $2^2 \times 3 \times 7$ **(d)** $2^2 \times 5 \times 7$

2. Which of the following is irrational? 1

(a) $\sqrt{\frac{4}{9}}$ **(b)** $\frac{\sqrt{12}}{\sqrt{3}}$ **(c)** π **(d)** 0.5

3. A student states: "The degree of the polynomial 7 is 1." The correct response is: 1

(a) Correct
(b) Its degree is 7
(c) Incorrect — a non-zero constant has degree 0
(d) It has no degree

4. A parabola $y = p(x)$ meets the x-axis at $(-2, 0)$ and $(5, 0)$. The zeroes of $p(x)$ are: 1

(a) $-2, 5$ **(b)** $2, -5$ **(c)** $-2, 0$ **(d)** $0, 5$

5. For $x + y = 8$ and $x - y = 2$, Student A gets $(5, 3)$ and Student B gets $(3, 5)$. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The value of k for which $2x + 3y = 5$ and $4x + ky = 10$ have infinitely many solutions is: 1

(a) 3 **(b)** −6 **(c)** 6 **(d)** 5

7. The discriminant of $2x^2 - 7x + 6 = 0$ is: 1

(a) 1 **(b)** 97 **(c)** −1 **(d)** 49

8. The roots of $x^2 + 5x = 0$ are: 1

(a) $5, -5$ **(b)** $0, -5$ **(c)** -5 only **(d)** $0, 5$

9. The common difference of the AP $0.6, 1.7, 2.8, \ldots$ is: 1

(a) 1.7 **(b)** 1.1 **(c)** −1.1 **(d)** 0.6

10. Which term of the AP $2, 6, 10, \ldots$ is 50? 1

(a) 14th **(b)** 25th **(c)** 12th **(d)** 13th

11. A 1.5 m boy casts a 2 m shadow. At the same time a pole casts a 10 m shadow. The height of the pole is: 1

(a) 5 m **(b)** 6 m **(c)** 13.3 m **(d)** 7.5 m

12. If $\triangle ABC \sim \triangle DEF$ with $\angle B = 70^\circ$ and $\angle C = 50^\circ$, then $\angle D$ is: 1

(a) 50° **(b)** 70° **(c)** 60° **(d)** 120°

13. If $\tan A = \frac{3}{4}$, then $\cos A$ is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{4}{3}$

14. The number of tangents to a circle that are parallel to a given secant is: 1

(a) 1 **(b)** 0 **(c)** infinitely many **(d)** 2

15. The arc length of a sector of radius 14 cm and angle 45° is: 1

(a) 22 cm **(b)** 11 cm **(c)** 5.5 cm **(d)** 44 cm

16. The area swept by a 7 cm minute hand in 15 minutes is: 1

(a) 38.5 cm² **(b)** 154 cm² **(c)** 11 cm² **(d)** 77 cm²

17. The curved surface area of a hemisphere of radius 14 cm is: 1

(a) 1232 cm² **(b)** 616 cm² **(c)** 1848 cm² **(d)** 2464 cm²

18. In $\bar{x} = a + h\frac{\Sigma f_i u_i}{\Sigma f_i}$, h is the: 1

(a) class mark **(b)** assumed mean **(c)** class width **(d)** total frequency

19. The empirical relation between mean, median and mode is: 1

(a) Median = 3 Mode − 2 Mean **(b)** Mode = 3 Median − 2 Mean

(c) Mean = 3 Median − 2 Mode **(d)** Mode = 2 Median − 3 Mean

20. The probability of drawing a non-face card from a deck of 52 cards is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{9}{13}$ **(c)** $\frac{3}{13}$ **(d)** $\frac{10}{13}$

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Can 12^n end with the digit 0 or 5 for any natural number n? Justify. 2

22. If α, β are the zeroes of $x^2 + 3x + 1$, find $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. 2

23. Find k if $x^2 - 2(k+1)x + k^2 = 0$ has equal roots. 2

24. In $\triangle ABC$, $DE \parallel BC$ with $AD = 3$ cm, $AB = 7.5$ cm and $EC = 3$ cm. Find AE. 2

25. Show that $(1, -1)$, $(-1, 1)$ and $(-\sqrt{3}, -\sqrt{3})$ are the vertices of an equilateral triangle. 2

26. If $\sin\theta = \frac{8}{17}$, find $\cos\theta$ and $\tan\theta$. 2

27. A ladder makes 45° with the ground and its foot is 4 m from the wall. Find the length of the ladder. 2

28. Tangents PA and PB are drawn from P to a circle. A third tangent touches the circle at C and meets PA at X and PB at Y. If $PA = 10$ cm, find the perimeter of $\triangle PXY$. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $x + 3y = 6$ and $2x + 6y = 12$. 3

30. Find the mean by the step-deviation method: 3

Class	0–100	100–200	200–300	300–400	400–500
Frequency	4	8	12	10	6

31. If $a_n = 9 - 5n$, show that the sequence is an AP and find S_{15}. 3

32. PQ is a chord of length 4.8 cm of a circle of radius 3 cm. The tangents at P and Q meet at T. Find TP. 3

33. In a circle of radius 6 cm, a sector has area 18.84 cm². Find its angle and arc length. (Use $\pi = 3.14$) 3

34. A bag has 6 red, 4 blue and some green balls. If $P(\text{green}) = \frac{1}{3}$, find the number of green balls. 3

35. A girl flies a kite with a 150 m string at 30° to the ground; a boy flies one with a 100 m string at 45°. Whose kite is higher and by how much? 3

OR

Two men on the same side of a 75 m tower observe its top at 30° and 60°. Find the distance between them. 3

36. Find the mode of: 3

Class	0–20	20–40	40–60	60–80	80–100
Frequency	5	8	13	7	7

OR

A student says the mode of the following data is 6. Verify. 3

Family size	1–3	3–5	5–7	7–9	9–11
Families	6	9	15	9	1

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** Four trees in a garden are at P(−1, 2), Q(3, 2), R(3, −1) and S(−1, −1). 5

(i) Find PQ and QR.
(ii) Find the diagonals PR and QS.
(iii) What type of quadrilateral is PQRS?
(iv) Find the centre of the plot.

OR

Find the point P on AB, where A(−2, 2) and B(2, 8), such that $AP = \frac{3}{7}AB$. Also find the mid-point of AB and its length. 5

38. Prove that $(\sin A + \cos A)(\tan A + \cot A) = \sec A + \csc A$. Verify it for $A = 45^\circ$. 5

OR

Prove that $(\csc\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$. 5

39. A rocket model is a cylinder of radius 3 cm and height 12 cm with a cone of radius 3 cm and height 4 cm on top. Find 5

(i) its volume
(ii) its total surface area.

OR

Case study: An oil drum is a cylinder of radius 0.7 m and height 2 m with a hemispherical top. 5

(i) Find its capacity in litres.
(ii) Find the curved area to be painted.
(iii) Find the cost of painting at ₹25 per m².

40. The median of the distribution is 32.5 and $\Sigma f = 40$. Find f_1 and f_2. **5**

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	f_1	5	9	12	f_2	3	2

OR

A student claims the mean of the data below is 40. **5**

(i) Find the actual mean.

(ii) Comment on the claim.

(iii) Find the median.

Class	0–20	20–40	40–60	60–80	80–100
Frequency	12	18	10	6	4

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND